WASP-35b

R-1507

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-35_190126` · created 2026-08-01T15:34:31+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458509.6804 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.124 +/- 0.013
Transit depth (Rp/Rs)^2	1.55 +/- 0.32 [%]
Orbital Inclination (inc)	89.03 +/- 1.26
Airmass coefficient 1 (a1)	0.9815 +/- 0.0082
Airmass coefficient 2 (a2)	0.013 +/- 0.014
Scatter in the residuals of the lightcurve fit is	0.84 %
Best Comparison Star	#1 - [195, 198]
Optimal Aperture	5.15
Optimal Annulus	7.53
Transit Duration (day)	0.1331 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.124 ± 0.013 vs 0.1241 (deviation 0.01σ)
Duration fit vs published	0.1331 d vs 0.1317 d (deviation 0.34σ)
Mid-time O–C	4.3 min
Depth SNR	9.5
Residual scatter	0.84 %

Light curve

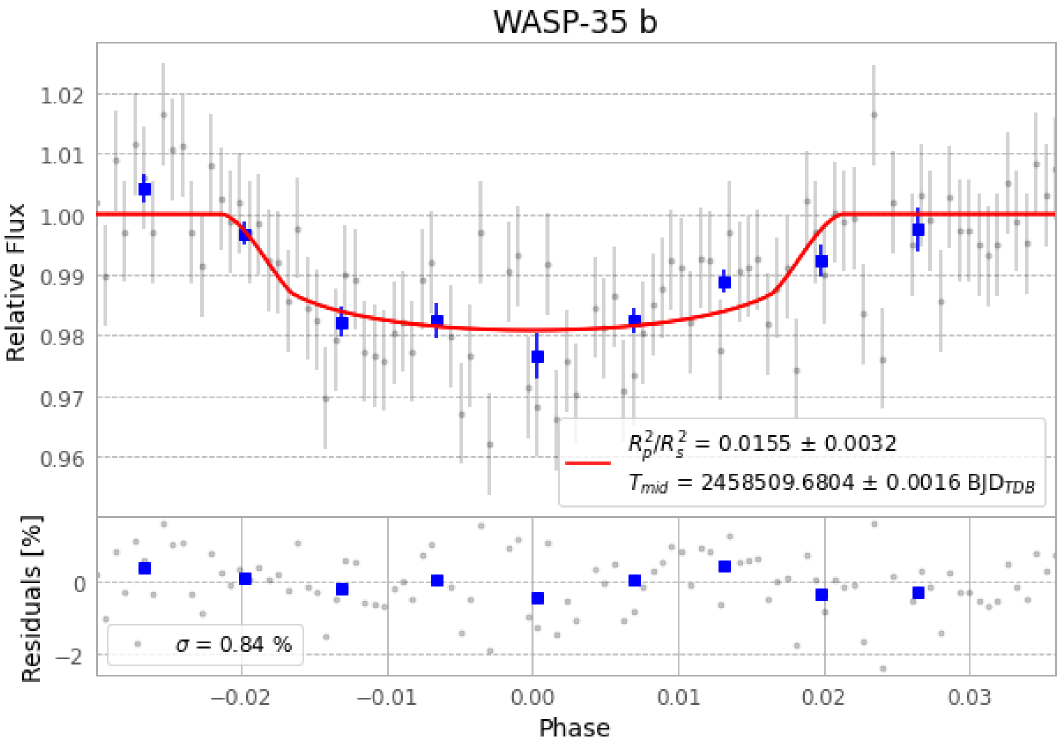

Figure 1 — FinalLightCurve_WASP-35 b_2019-01-25.png (SHA-256 855990e459ed103a…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [380, 139], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 8d94fdf6c3d4099f…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

100 FITS frames (66 MB), `WASP-35190126015712.FITS` → `WASP-35190126065712.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-190126.log (ac1a947b1486…), FinalLightCurve_WASP-35 b_2019-01-25.png (855990e459ed…), FinalLightCurve_WASP-35 b_2019-01-25.csv (d02a76b92152…)

Compute resources

Wall time 656.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)